

Feasibility Study: Technical and Economic Assessment of a New Wind Farm Project in a Mountainous Region in Central Sweden

Team 2

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ABSTRACT

In Sweden, wind energy is crucial to the national strategy for decarbonization and fulfilling the Paris agreement. This project aims at evaluating technical and economical feasibility of an 80 MW wind farm project in a mountainous region in Sweden. In the first part, turbine technology is explored. After comparing different turbine types and deciding on a suitable mountainous site close to Idre fjäll in bidding area SE3, different topological and electrical layouts are assessed. The chosen turbine is the Vestas V136-3.45 turbine, in a staggered formation and with an open-ring electrical layout. Next, different grid connections are explored and the cost of connection is calculated. The net present value and levelized cost of energy are calculated for a range of different investment costs. It is shown that when using the lower boundary investment cost, the project is profitable, but when using the upper boundary, it is not. Lastly, a hybrid battery energy storage (BESS) system is investigated, and it is shown that the total revenue increases when adding a battery. The BESS system is evaluated for different investment costs, resulting in different optimal size of the battery. The overall feasibility is dependent on different economic and technical assumptions, which is discussed in the end of the report.

Index Terms—Wind power, feasibility study, turbine technology, grid connection, LCOE, BESS.

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I. INTRODUCTION

The accelerating global energy transition places renewable energy at the heart of international decarbonization strategies. According to the International Renewable Energy Agency (IRENA), renewable sources could supply more than 60 % of global electricity by 2050, with wind and solar providing the largest contributions [1]. Pathways aligned with the Paris Agreement [2] emphasize that wind power will have a leading role in achieving climate neutrality. Over the past decade, wind technology has matured significantly, with large-scale deployment supported by declining costs, improved turbine efficiency, and digital optimization tools [3].

More specifically, the global weighted average Levelized Cost of Electricity (LCOE) for onshore wind has fallen by over 56 % between 2010 and 2022, reaching around 0.033 USD/kWh [4]. This cost competitiveness has made wind energy one of the most viable large-scale renewable technologies, capable of displacing fossil-based generation in both industrialized and developing energy systems [4]. However, as the penetration of the variability of renewable energy increases, power systems face new operational challenges: balancing supply and demand in real time, ensuring voltage and frequency stability, and providing sufficient system flexibility.

Within Europe, European Commission through the European Green Deal has set ambitious targets for reducing greenhouse gas emissions by 55 % by 2030 and achieving a fully decarbonized power sector by 2050 [5]. Europe now has 285 GW of wind power capacity, and wind power is expected to become the largest source of electricity generation in Europe within the next decade [6].

In the Nordic region, and particularly in Sweden, wind energy is central to the national decarbonization roadmap. The Swedish government has committed to achieving 100% renewable electricity production by 2040 [7]. In addition, wind generation is projected to exceed 45 TWh by 2030, supported by strong policy incentives and regional grid expansion. Wind power complements the country's dominant hydropower resources by offering seasonal and geographical diversification and strengthening overall system flexibility. Nevertheless, many of the remaining high-potential onshore sites are located in mountainous and forested regions—areas characterized by complex terrain, cold climates, and limited accessibility. Developing wind farms in such environments requires specialized turbine design, optimized layouts and robust grid-integration solutions to ensure efficient and reliable operation.

Within this broader framework, the present work aims to evaluate the technical and economic feasibility of an 80 MW onshore wind farm in a mountainous Swedish site. The study, conducted by Team 2 as part of the KTH course EG2340 - Wind Power Systems (HT25), follows a structured methodology defined in the official project assignment [8]. The report is organized into six main tasks, each addressing a specific aspect of the design and assessment process:

- Task A—Turbine Technology Analysis: Review and compare four commercial wind-turbine models (1-8 MW) from different manufacturers in terms of rated power,

rotor diameter, generator type, power factor, and grid-support functions. Two representative turbines are selected for further analysis.

- Task B—Siting & Wind Data Analysis: Identify a suitable mountainous site in Sweden, justify the location choice (wind resource, grid access, and environmental constraints), and design two alternative layouts (topological and electrical). Using data from Renewables.ninja and the Global Wind Atlas, the Annual Energy Production (AEP) and capacity factors are estimated with wake-effect modeling.
- Task C—Grid Integration: Evaluate potential connection points and voltage levels, assess reactive-power capability, and analyze the impact of turbine characteristics on grid stability.
- Task D—Economic Analysis: Compute the Net Present Value (NPV) and Levelized Cost of Electricity (LCOE) for the candidate designs over a 25-year lifetime, accounting for capital and operational expenditures as well as energy yield.
- Task E—Battery Energy Storage System (BESS): Assess the potential of storage integration to improve flexibility, mitigate curtailment, and enhance economic performance.
- Task F—Overall Feasibility & Recommendation: Compare all configurations and recommend the optimal turbine-layout combination based on technical and economic indicators.

By integrating turbine technology evaluation, wind-resource assessment, grid-integration analysis, and economic modeling, this project provides a holistic understanding of how site conditions and system requirements influence the feasibility of modern onshore wind-farm developments in complex terrain. The results aim to illustrate the trade-offs between cost, energy yield, and grid support, contributing to the broader goal of expanding Sweden's renewable energy capacity toward its 2040 target [7].

II. WIND TURBINE TECHNOLOGY

In the first part of this section, four different wind turbine technologies are described. From these technologies, two are chosen as superior for a comparison of on-the-market wind turbines. Four turbines of these types from different manufacturers are then compared for rated power, rated wind speed, size, available packages, foundation types, and more. Based on the comparison, two of the turbines are chosen for further analysis in later sections.

A. Type A: Danish Concept

The Danish concept is a fixed-speed turbine which rotates with the same speed independent from the wind. This means the rotor runs at nearly constant rotational speed, constrained by the synchronous speed of the generator rather than the wind. This results in a relatively simple design. The blades have a fixed angle that is rotating a low-speed shaft. The shaft is coupled to a gearbox with a fixed gear ratio transferring the rotational speed to a high-speed shaft which is linked to an asynchronous (or induction) generator directly connected

to the grid. Therefore, the gear-ratio as well as the number of pole pairs sets the speed of the blades. For example, the turbine is connected to a 50 Hz grid and has a gearbox with gear ratio of 1:50 and the generator has 4 pole pairs. This results in a rotational speed of the blades of:

$$\frac{60 \cdot 50}{3} \cdot \frac{1}{50} = 20 \text{ rpm} \quad (1)$$

Because it is a fixed-speed turbine, it will always rotate at around 20 rpm, varying by only a few percent due to slip. The output power from the turbine depends strongly on the difference between the line frequency and the generator's rotational speed. Maximum power is extracted when the rotor speed is a few percent above the synchronous speed [9]. Therefore, it is unlikely to extract the maximum aerodynamic power from the wind, resulting in an inefficient turbine design.

The control of the turbine at high wind speeds is achieved through stall regulation, because the blades have a fixed pitch angle. When the wind speed increases above rated, the angle of attack also increases, which reduces lift and caps the aerodynamic torque [10].

B. Type B: Variable Resistance

For fixed-speed wind turbines, the rotor speed is fixed by the system frequency, and cannot adjust to changes in the wind speed. This causes grid integration issues in power systems where wind power is prominent, especially in combination with weak supply grids [11]. Moreover, the power coefficient and the aerodynamic power is limited [12]. To account for these problems, variable-speed wind turbines were developed. Small variable-speed wind turbines were introduced in the US and elsewhere in the 1920s, with typical installations ranging between 0.5-10 kW [11].

The idea of the variable resistance wind turbine is to add an external variable resistance to the rotor of an induction machine [12], typically a wound rotor induction generator (WRIG) [10]. The external rotor resistance enables increased generator operating speed at a given torque, effectively increasing the rotating speed. Increasing the rotor speed increases the tip speed ratio λ . This is seen from the relationship between the tip speed λ , the rotor radius R , the rotor speed ω_r and the wind speed v [12]:

$$\lambda = \frac{R\omega_r}{v} \quad (2)$$

The power coefficient C_p is dependent on the tip speed ratio, which must be kept near its optimal value in order to maximize the extracted energy of the turbine rotor. This is achieved by adjusting the rotor speed with the varying wind speed [10], as seen in Eq. (2). The variable resistor controls the tip speed to keep the output power as high as possible. This leads to an increased power coefficient and aerodynamic power compared to a fixed-speed wind turbine, up to the cut-out speed where the turbine is stalled for protection [12].

Another advantage of the variable-resistance application compared to the fixed-speed turbine is that it does not require pitch control alone to account for wind speed variations. This enables faster control and reduces the wear on the pitch control

system. Instead, the variable resistance control and the pitch control can be combined to increase power output [12].

The major disadvantage of the variable resistance turbine is that the increased rotor resistance leads to more power losses. While the increased aerodynamic power gain is larger than the increased losses [12], there are other applications than the WRIG to account for this. One advantage of using a WRIG compared to the applications discussed in later sections is that the variable-speed operation is achieved using only 1/3 the amount of power converters [10], effectively decreasing component cost.

C. Type C: DFIG

DFIG wind turbines use a doubly fed induction generator to obtain variable speed operation while keeping the power electronics only partially rated [13]. The machine is a wound rotor induction generator whose stator is connected directly to the grid at line frequency, and whose rotor is connected through a bidirectional back to back converter that typically handles on the order of one third of the rated power [14]. By imposing appropriate rotor currents, the converter controls the slip so that the stator electrical frequency remains fixed while the mechanical speed varies around synchronous speed [15]. Power can flow in both directions through the rotor circuit, which enables subsynchronous operation with power drawn from the grid into the rotor and supersynchronous operation with rotor power returned to the grid [13] [15]. In practice the controllable speed range is commonly designed to about plus or minus thirty percent so that the rotor side converter can be dimensioned to roughly thirty percent of generator rating, although wider ranges are achievable when the converter is sized accordingly [14] [15].

The control system regulates electromagnetic torque through vector control of the rotor current, which provides fast adjustment of active power and thereby of rotor speed. By adjusting the phase of the rotor current, the machine also controls its power factor, which means it can supply or absorb reactive power as required by the grid. In effect the DFIG combines the asynchronous robustness of an induction generator with many grid support capabilities familiar from synchronous machines, while only a fraction of the power passes through the converters [13] [15].

The principal advantages follow from this structure and control. Variable speed allows the rotor to remain close to the aerodynamic optimum over a range of wind velocities, which increases annual energy production relative to fixed speed concepts and reduces mechanical fatigue through smoother torque behavior [13]. The ability to shape electromagnetic torque and to provide reactive power improves grid integration at the point of connection, while converter based synchronization enables soft and well controlled grid connection without dedicated soft starters [14]. Because only a fraction of the power is processed by the converter, electrical losses and converter cost remain comparatively low [13].

The main disadvantages arise from the direct coupling of the stator to the grid and from additional hardware in the rotor circuit. Slip rings and brushes are required to feed the

rotor, which introduces wear, periodic maintenance, and an additional failure mode [15]. During deep voltage dips in the grid the stator experiences large transients that couple into the rotor, and the partially rated converter may not withstand the induced currents, so protective measures such as a crowbar are required and during such events reactive power control is temporarily lost [14]. Upfront costs are higher than for fixed speed or variable resistance induction concepts because of the converter and controls, although they remain below those of full converter machines [14]. The usual presence of a multi stage gearbox maintains the maintenance effort associated with geared designs.

When positioned among the four canonical types, the DFIG constitutes a middle ground. Relative to the Danish concept with a squirrel cage induction generator at fixed speed, the DFIG offers higher energy yield and markedly better grid support, since the fixed speed concept depends on capacitor banks for excitation and exhibits steep torque speed characteristics that can cause flicker and elevated mechanical stresses [14]. A wound rotor machine with variable rotor resistance extends the speed range only modestly and continues to dissipate slip power as heat, which limits efficiency gains and leaves reactive power control largely unaddressed [14]. In contrast, a full converter machine processes all power through its converters and thereby decouples the generator from the grid, which enables a much wider speed range including very low speed operation and supports superior fault ride through as well as the option of direct drive without a gearbox [14] [16].

D. Type D: Full converter

Type D wind turbines, also called Type IV, use a full scale back to back power electronic converter between the generator and the grid. The generator is completely separated from grid frequency. In practice, this design is often used with multi pole synchronous machines. These can be electrically excited or use permanent magnets. It can also be used with induction machines. The converter changes the generator output to direct current, then to alternating current at grid frequency. This allows continuous variable speed operation. The rotor can stay near the optimal tip speed ratio and increase annual energy capture under changing winds [17].

Full converters give strong grid integration and flexible control. Active and reactive power can be controlled independently. Voltage can be regulated at the point of common coupling. Modern systems meet strict grid code rules for power quality and fault ride through. During voltage dips, the converter can control currents to protect the machine and ride through faults while supporting reactive power. This ability is more limited in Type C designs where the stator is tied to the grid [17][18]. The fast electronic separation also enables ramp rate limiting and output smoothing, which is useful on weak or remote grids [18].

In mechanics and reliability, Type D lowers drivetrain stress. The rotor can speed up or slow down with gusts instead of holding a fixed speed. When used with permanent magnet direct drive, the concept removes slip rings, brushes, and the gearbox. This can cut maintenance needs and electrical

failures compared with earlier designs. It is helpful at sites with complex terrain where access is hard and winds are very turbulent [17][18].

The main tradeoffs are cost, complexity, and losses in a full rated converter sized for 100 percent of turbine power, compared with the partially rated converter in Type C machines. Large direct drive generators can be heavy. Permanent magnet designs depend on rare earth materials, which can raise nacelle mass and upfront cost. Full converter turbines also supply limited short circuit current, because the converter limits fault current to near rated levels. This can make protection harder on very weak systems. Even so, for most modern onshore projects, especially at sites with variable winds or weak grids, the higher energy capture, better control, and strong grid code compliance of Type D usually outweigh these drawbacks [17][18][19].

E. Different turbine models

In this subsection, four different turbines from different manufacturers are compared in terms of factors such as rated power, rated wind speed, size, and more. Medium-sized turbines of Type c (DFIG) and Type D (Full converter) from prominent European and Chinese manufacturers were chosen, and turbine specifications of the chosen turbines are presented in Table I. These specifications were found from [20][21][22] for the Vestas turbine, [23][24][25] for the Enercon turbine, [26][27] for the Siemens-Gamesa turbine, and [28][29] for the Envision turbine. The scope of which turbines to choose were delimited to Type C and D because they represent the most advanced and grid-compatible technologies currently available for onshore wind projects. In addition, medium-sized turbines (between 3-5 MW of rated power) were chosen as a trade-off between large turbines and the need for many turbines to account for limitations within the mountainous area.

From the turbines presented in Table I, two are chosen to be best suited for our wind project. The Vestas V136-3.45 and the Enercon E138/4200 stand out as the best options for a wind farm in a mountainous area. The five factors considered in the comparison are the following.

1) *Wind Class*: Assuming that a mountainous area is characterized by medium wind conditions with high turbulence, wind turbines rated for the IEC wind class IIA is best suited [30]. The mountain area is likely similar to on-shore conditions but with higher turbulence due to the rough terrain of mountains. Interacting with rough landscape such as ridges, valleys and varying elevation causes the wind to be more turbulent. Both the Vestas V136-3.45 and Enercon E138/4200 are classified as IIA, which is the medium-wind high-turbulence wind class [30]. The Siemens SG 5.0-145, with an IIB rating, is also medium-wind but does not account for higher turbulence. The Envision EN156-3.3 is rated only for Class S, making it the least suited for complex terrain and turbulence.

2) *Turbine Size*: In mountainous areas, the transport and installation of large turbines can become a challenge due to slope and rough terrain. The Vestas V136-3.45 and the Enercon E138/4200 have rotor diameters of 136m and 138m

Table I
COMPARISON OF CANDIDATE WIND TURBINE MODELS FOR THE 80 MW MOUNTAIN ONSHORE PROJECT

Parameter	Vestas V136-3.45	Enercon E138/4200	Siemens SG 5.0-145	Envision EN156-3.3
Rated Power [MW]	3.45	4.2	5	3.3
Hub Height [m]	82, 105, 112, 132, 142	81, 91, 111, 131, 160	90, 102.5, 127.5	140.5
Rotor Diameter [m]	136	138	145	156
Generator / Type	Type IV (Full Conv.)	Type IV (Full Conv.)	Type III (DFIG)	Type III (DFIG)
Cut-in Speed [m/s]	3	2.5	3	3
Rated Speed [m/s]	10	10.8	11.2	9.6
Cut-out Speed [m/s]	22.5	28	27	25
Power Factor Range	-	-	± 0.9	-
Fault Ride-Through Capability	High	High	Medium	Medium
IEC Wind Class	IIA	IIA, IIIA, S	IIB	S
Sound [dB(A)]	≤ 105.5	99-106	102.7-106.3	-
Cold-Climate Package	Yes	Yes	Yes	No
Foundation Type	Custom to site	Custom to site	Custom to site	-

respectively, which is relatively compact compared to the larger Siemens SG 5.0-145 (145 m) and Envision EN156-3.3 (156 m). This makes the Vestas and Enercon turbines easier to transport and install in challenging mountain terrain. Moreover, the Vestas V136-3.45 and Enercon E138/4200 offer different hub height options, ranging from 81 m to 160 m. While the Siemens SG 5.0-145 also offers different hub heights, its smallest size is 90 m. The Envision EN156-3.3 has a fixed hub height of 140.5 m, which could limit its adaptability to varying mountain elevations.

3) *Turbine Technology*: Both the Vestas V136-3.45 and the Enercon E138/4200 turbines have Type D (Full converter) generators, which are more efficient and reliable for variable wind conditions. Moreover, these turbines do not have a direct connection to the grid, which is beneficial for grid integration. The Siemens SG 5.0-145 and Envision EN156-3.3 have Type C (DFIG) generators, which offer less flexibility and are in part directly connected to the grid.

4) *Cold-Climate Package*: At higher altitudes, it is likely that there will be cold temperatures, especially during the winter. The Vestas V136-3.45, Enercon E138/4200, and Siemens SG 5.0-145 all offer cold-climate packages. However, the Envision EN156-3.3 specification does not specifically state that it has this feature, which may be beneficial in the mountain location.

5) *Sound*: Quieter wind turbines are preferred over noisier turbines. The Enercon E138/4200 is the quietest of the options, with sound levels ranging from 99-106 dB(A). The Vestas V136-3.45 has a slightly higher sound level of 105.5 dB(A). The Siemens SG 5.0-145 is slightly noisier, with sound levels between 102.7-106.3 dB(A).

III. SITING AND WIND DATA ANALYSIS

In this section, a location is selected for the wind turbine site. Based on the conditions of this site, two graphical layouts are created for both topological and electrical layout. Then the annual energy production and capacity factor for the two wind turbines chosen in the previous section is calculated for both layouts. Lastly, the scope is narrowed down to one turbine and one layout.

A. Location of site

Since mountainous regions in Sweden are often desolate areas far from electricity demand and grid connection, the feasibility study for the location of the site for the wind farm is based on real wind farm projects under development. This ensures that the site is suitable for a wind farm project in terms of elevation variations, grid connection possibilities, proximity to electricity demand and infrastructure. The potential of the site is still rather unknown, since it is not yet a realized project. It is arguably interesting to use the results of the pre-study of the planned real-life project as a benchmark to compare to the results of our study.

The criteria used for the candidate sites were that it should have a planned wind farm that is currently under development in a mountainous area in Sweden. Moreover, wind conditions, economic factors, and possible grid connection were considered. Based on these criteria, the site Skärvagshållan was chosen as our wind farm project site. The planned real-life wind farm has a rated power of 98 MW which is relatively similar to our project, which enables comparability. The plan for the real-life project includes 13 wind turbines with a hub height of up to 290 m [31], and an annual energy production of 346 GWh [32].

Skärvagshållan is located 20 kilometers west of Idre, near Idre fjäll. The site is positioned at a height of 700 m above sea level [33]. The highland conditions in Swedish fell areas, combined with its elevation above sea level, classify the site as mountainous. The wind conditions are also reasonable in relation to our estimation of wind class. We chose turbines rated for the IEC wind class IIA, which corresponds to a wind speed of 8.5 m/s and high turbulence [30]. The wind conditions at the site at height 100 m is 8.78 m/s [34].

This location also ensures economic feasibility, as it belongs to the bidding area SE3 Stockholm [32]. Compared to the more remote mountainous fell areas in Northern Sweden, this site is located in the central part of Sweden where the electricity demand is significantly higher. Being closer to consumption centers reduces transmission bottlenecks. It also allows energy to be sold at higher spot-market prices, increasing the potential profitability of the wind farm project.

Moreover, there is an ongoing project from the network owner Ellevio for a new 150 kV transmission line and substation in Brattströmmen, with construction expected to begin in 2026 [35] [36]. The new 150 kV Brattströmmen substation will be located approximately 30 kilometers from our site [32], which shows good prospects for future grid connections and sufficient capacity to connect an 80 MW wind farm.

B. Wind data

The wind data at the coordinates of our sites was downloaded from renewables.ninja [37]. The data includes hourly time series of wind speed between the years 2021-2024 at 10 m height. We also retrieved data for the Vestas V136-3.45 turbine of the electricity in kW at hub height, to compare our calculated values of annual energy production to those from the data set.

To upscale the wind speed to the hub height of approximately 130 m, the following equation is used, where z is the hub height, $U(z)$ is the wind speed at hub height, z_r is the reference height, $U(z_r)$ is the wind speed at the reference height, and α is the power law exponent:

$$U(z) = U(z_r) \left(\frac{z}{z_r} \right)^\alpha \quad (3)$$

To find a suitable value of α , a different data source is used to estimate it based on wind speed values for different heights. By rearranging the formula in Eq. (3), the following expression for the power law exponent is found:

$$\alpha = \frac{\ln \left(\frac{U(z)}{U(z_r)} \right)}{\ln \left(\frac{z}{z_r} \right)} \quad (4)$$

Single values of wind speeds at 10 m and 100 m respectively from Global Wind Atlas [34] are used to estimate α . This yields a value of α of 0.43. This is a significantly higher value than what would be found from estimating it from the surface roughness length, but it can likely be attributed to local terrain influences that cause increased shear in the vertical wind profile.

A wind rose showing the distribution of wind directions is retrieved from the Global Wind Atlas [34]. The values of the wind rose is listed in Table II.

From these values, a mean wind direction is calculated. With i denoting the section in the wind rose, θ_i denoting the wind direction of a section, and f_i denoting the relative frequency of a section, the mean wind direction $\bar{\theta}$ can be calculated using the formula for the circular mean [38]:

$$\bar{\theta} = \text{atan2} \left(\sum_i f_i \sin \theta_i, \sum_i f_i \cos \theta_i \right) \quad (5)$$

The mean wind direction at our site is calculated to be 222.71° , with clockwise rotation from North. This corresponds approximately to a South-west direction. The mean wind direction is used as a reference for the topological layout in following sections.

Table II
WIND DIRECTION DISTRIBUTION

Section	Wind direction ($^\circ$)	Relative Frequency
1	0	0.06
2	30	0.04
3	60	0.04
4	90	0.05
5	120	0.05
6	150	0.06
7	180	0.14
8	210	0.21
9	240	0.09
10	270	0.05
11	300	0.07
12	330	0.12

C. Methodology for AEP and CF calculation

Based on the wind dataset introduced earlier, a systematic procedure was applied to estimate the Annual Energy Production (AEP), Capacity Factor (CF), and wake losses for the candidate layouts and turbine models. The analysis uses the upscaled hourly wind-speed measurements at hub height of 130 m height, which is applicable for the both wind turbines.

Directional statistics were discretized into twelve directional sectors of 30° each and each sector was assigned a probability of occurrence p_k (Tab. II). For each sector, layout and turbine configuration, the farm energy yield was computed and then weighted using the next formula:

$$\text{AEP} = \sum_{k=1}^{12} p_k \cdot \text{AEP}_k, \quad (6)$$

where AEP_k represents the accumulated hourly power output of all turbines in the wind farm, for all timestamps under sector k .

The capacity factor is obtained from the rated power $P_{\text{rated,tot}}$:

$$\text{CF} = \frac{\text{AEP}}{P_{\text{rated,tot}} \cdot 8760} \quad (7)$$

An essential criterion for the proper evaluation of AEP and CF is the power conversion from wind speed to electrical output which is performed via turbine power curves. For the Vestas V136-3.45 [22], different operational points from the theoretical manufacturer power curve were available and by using linear interpolation, the theoretical rated behaviour was modeled. For the Enercon E-138/4200 [23], no instantaneous power curve is publicly available. Only annual yield versus mean speed is provided, which is not useful for assessing instantaneous power output for different wind speeds. To allow comparison between the two turbine candidates, the Enercon power curve was approximated using a cubic function between cut-in and rated wind speed. This simplified representation cannot reproduce the actual pitch-regulation strategy and torque control of the turbine, and therefore introduces systematic modelling bias, potentially resulting in over- or under-estimation of AEP and capacity factor.

Wake interactions were modeled with a Jensen/Park-type approach. The wake expands linearly downstream, and the velocity deficit depends on the distance between turbines, the rotor diameter D , and a terrain-dependent wake decay constant $k \approx 0.075$. For every hourly timestamp and directional sector, the model identifies whether a downstream turbine is within the wake cone of an upstream one. The local reduced wind speed at each turbine $u_{i,t}$ is computed as:

$$u_{i,t} = u_{0,t} \left(1 - \sqrt{\sum_j \Delta u_{j \rightarrow i,t}^2} \right), \quad (8)$$

where $u_{0,t}$ is the undisturbed speed and $\Delta u_{j \rightarrow i,t}$ is the deficit caused by turbine j . The implementation includes interpolation of the thrust coefficient $C_t(u)$ from a tabulated dataset, geometric wake inclusion, and superposition of deficits.

D. Topological layout

The geometric configuration of the wind farm strongly influences wake losses and therefore the annual energy production (AEP). Industry guidelines suggest that streamwise spacing between turbines should be between $6D$ and $8D$, while lateral spacing typically ranges from $3D$ to $5D$ (with D denoting the rotor diameter), depending on terrain complexity and wake sensitivity [10]. Following these recommendations, a streamwise separation of approximately $7D$ and a lateral separation of $4D$ were adopted.

Two alternative topologies were evaluated:

- 1) Staggered / offset layout: In this configuration, turbines are arranged in two rows such that each turbine in the downstream row is laterally shifted relative to the one upstream. Such arrangements reduce direct wake overlap and are widely used in flat-terrain commercial wind farms.
- 2) Arced ridge-aligned layout: The second configuration positions turbines along an arc oriented with the dominant wind direction $\approx 222.7^\circ$, as identified from the wind rose analysis. Ridge-aligned layouts are recommended for complex mountainous terrain, where local speed-up effects near mountain crests may increase wind velocity and reduce wake interaction along the flow direction [39]. Aligning the turbines along the prevailing direction reduces along-flow wake stacking, which is particularly beneficial when the site's dominant wind direction is persistent. In this study, two alternative inter-turbine spacings were evaluated for the arced ridge-aligned configuration. The turbines are arranged along a circular arc, with a constant centre-to-centre distance of either $7D$ (952 m) or $4D$ (544 m) between neighbouring machines.

For each of the twelve directional wind sectors, the corresponding layout is rotated mathematically to align with the incoming wind direction. Wake-induced velocity deficits are computed via the Jensen/Park model, turbine power curves are applied to obtain the instantaneous electrical output, and

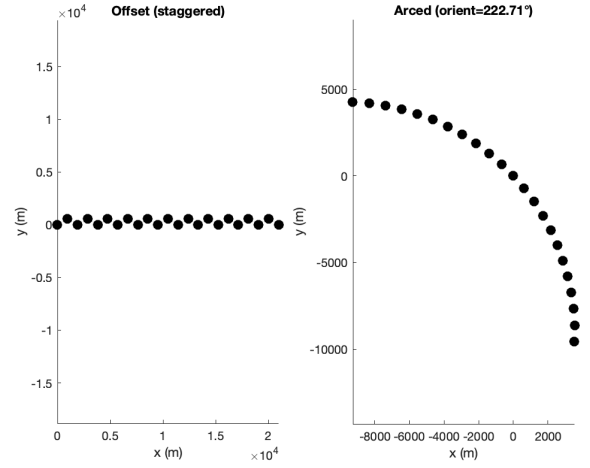


Figure 1. Topological layouts evaluated in this study: (left) offset (staggered) arrangement and (right) arced ridge-aligned layout oriented to the dominant wind direction (222.71°).

the final AEP is computed using the probabilistic weighting of Eq. (6).

This procedure enables a physically consistent comparison of the two layouts with respect to wake losses and energy production, while capturing both directional variability and topographic influence.

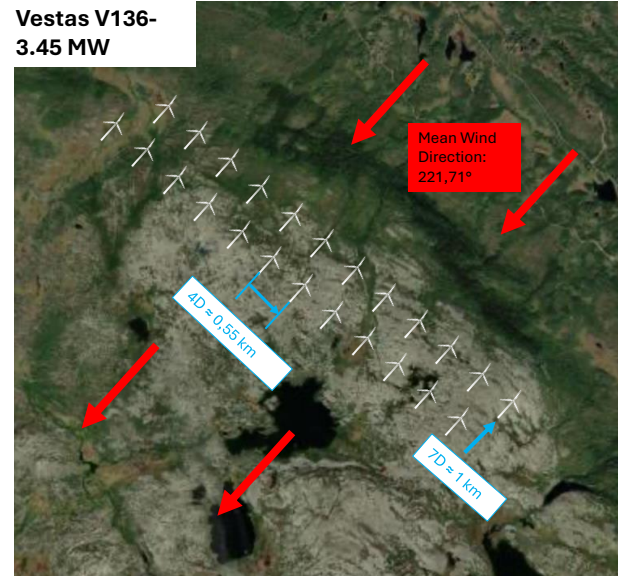


Figure 2. Schematic two-row staggered topology with 24 Vestas V136–3.45 MW turbines (Enercon E–138/4.2 MW would need only 19 turbines).

E. Comparison of AEP, capacity factor and wake losses

Based on the aforementioned methodology, the annual energy production (AEP), capacity factor (CF) and wake losses were computed for both turbines (Vestas V136–3.45 MW and Enercon E–138/4.26 MW) and for both proposed topological layouts. For the arced ridge-aligned layout two different spacing proposals are evaluated. The dataset from *Renewables.ninja* was used as an external reference to assess the

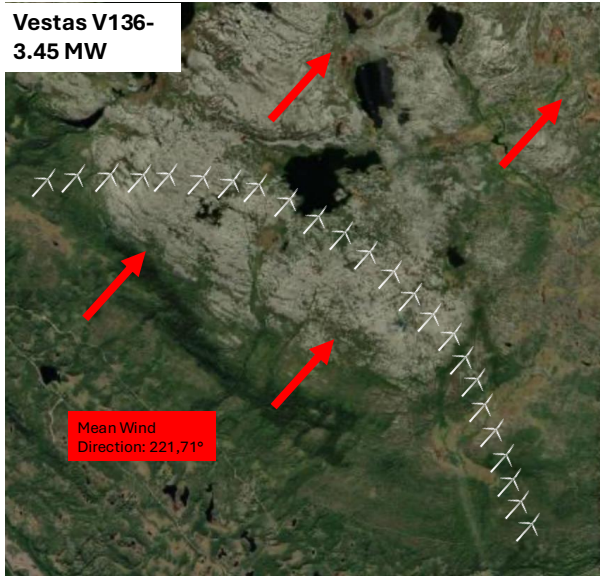


Figure 3. Schematic arced ridge-aligned topology with 24 Vestas V136–3.45 MW turbines (Enercon E–138/4.2 MW would need only 19 turbines).

accuracy of the results, since it provides long-term energy yield simulations.

Table III
AEP, CAPACITY FACTOR AND WAKE LOSSES FOR ALL EVALUATED LAYOUTS AND BOTH TURBINE MODELS.

Turbine	Layout	AEP [MWh/yr]	CF [%]	Wake Loss [%]
Vestas V136	Staggered	270297.6	38.89	2.91
	Arced (7D)	273064.4	39.28	1.92
	Arced (4D)	267181.2	38.44	4.03
Enercon E138	Staggered	282092.4	39.79	1.98
	Arced (7D)	284228.9	40.09	1.24
	Arced (4D)	279248.2	39.38	2.97

Regarding the comparison between the examined layouts, the arced configuration with 7D spacing yields slightly higher AEP and reduced wake losses compared with the proposed staggered placement. However, despite its aerodynamic advantages, the 7D arced configuration exceeds the spatial limits of the studied site. As illustrated in Fig. 1 and Fig. 3, this layout requires an approximate horizontal extent of 13.9 km and a vertical extent of about 12.8 km for the full arc to be accommodated, which is significantly larger than the available examined area. For this reason, although the 7D arced case demonstrates favourable wake characteristics, it is not feasible within the boundaries of the investigated terrain.

To address this limitation, a reduced-spacing variant using 4D separation along the arc was also evaluated. While this compact version fits within the spatial constraints of the site, its performance is not as competitive: in both turbine models (Vestas V136 and Enercon E138), the 4D arced layout results in lower AEP and notably higher wake losses compared with the staggered layout. Consequently, although ridge-aligned geometries remain aerodynamically attractive, the compact 4D arc does not outperform the baseline staggered configuration

for the given site characteristics. As a result the proposed staggered option will be used and examined in this study.

Table IV
VALIDATION OF MODEL-DERIVED CAPACITY FACTOR AGAINST RENEWABLES.NINJA BENCHMARK.

Wind Turbine	CF _{model} [%]	CF _{ninja} [%]
Vestas V136	38.89	38.78
Enercon E138	39.74	34.05

The computed AEP and CF for the Vestas V136 show excellent agreement with the independent benchmark, with deviations within approximately 1 %. This confirms that the use of the manufacturer power curve combined with the wake-loss model leads to reliable energy estimates. In contrast, the Enercon E–138 in all layout proposals, systematically overestimates AEP and CF relative to the same benchmark. This deviation originates probably originates from the cubic approximation used to emulate the turbine power curve, which does not capture partial-load control behaviour nor the real shape of the power curve.

Considering:

- The availability of the true manufacturer power curve for Vestas.
- The validation of the Vestas results against an independent external benchmark.
- The higher AEP, lower wake losses, and overall suitability of the staggered configuration within the examined area

The project proceeds with the Vestas V136–3.45 MW turbine combined with the proposed staggered layout for the electrical and economic optimization stages.

All MATLAB scripts and data files used in this analysis are version-controlled in the project repository (GitHub), ensuring full reproducibility and traceability of the results.

F. Electrical layout

In both proposed electrical layouts the wind turbines are grouped in rows along the ridge. Each turbine operates at 690 V and is connected to an individual LV/MV step-up transformer. The turbines in one row are connected to a medium-voltage (MV) collector feeder. All feeders terminate at the on-site substation, where the power is stepped up to the transmission level by the main MV/150 kV transformer and then exported to the strong grid through the 150 kV line and a 150/400 kV grid transformer. The two layouts differ in the configuration of the MV collection system between the turbines and the substation.

In the first layout, the radial collectors, each feeder is a simple radial line that runs directly from the substation to a string of turbines. Several turbines are connected in series along one MV cable, and there is only one electrical path from each turbine to the substation and to the external grid. A fault on a cable, switch or transformer along a feeder disconnects all turbines located behind the fault until repairs are completed [40].

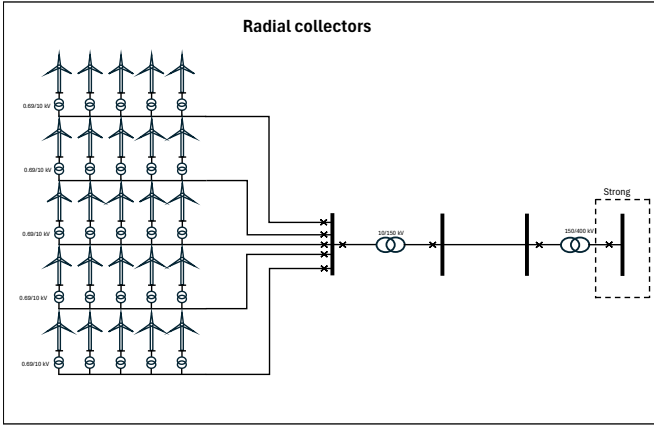


Figure 4. Electrical layout of the radial collectors

The main advantage of the radial layout is its simplicity. The configuration is easy to design and operate, the power flow on each feeder is always unidirectional towards the substation, and the protection settings are straightforward. Because the topology uses a relatively small amount of MV cable and a limited number of switching devices, the initial investment cost is also lower compared with more redundant solutions [40].

The main disadvantage of the radial layout is its limited reliability. Since there is no alternative path, a single component failure can stop the production of a whole string of turbines, especially when feeders are long and contain many machines. This results in higher expected energy not produced and offers little flexibility for planned maintenance, because taking one feeder out of service normally means that the connected turbines must be shut down [40].

In the second layout, the open-ring radial collectors, the turbines are still arranged in strings that connect radially to the substation, but the remote ends of the MV feeders are additionally interconnected. Sectionalising switches are installed so that the feeders form a ring, which is normally operated with one open point. Under normal operating conditions the system behaves like the radial layout, but if a fault occurs on one cable section or switch, the faulty part can be isolated and the open point can be moved so that healthy sections are re-energized from the opposite side of the ring [40].

The open-ring layout has the advantage of higher reliability and availability. Because there is an alternative electrical path, fewer turbines are disconnected for a single MV fault, and more turbines can remain in service during cable or switch maintenance. This reduces the loss of production for both unplanned faults and planned outages and can increase the total energy yield over the project lifetime [40].

The disadvantages of the open-ring layout are related to cost and complexity. The interconnection between feeders requires additional MV cables, more switches and a more advanced protection and control system. Power flows in the ring can become bidirectional after reconfiguration, and the position of the open point may change, which makes the protection concept more involved. As a result, the capital expenditure and engineering effort are higher than for the simple radial

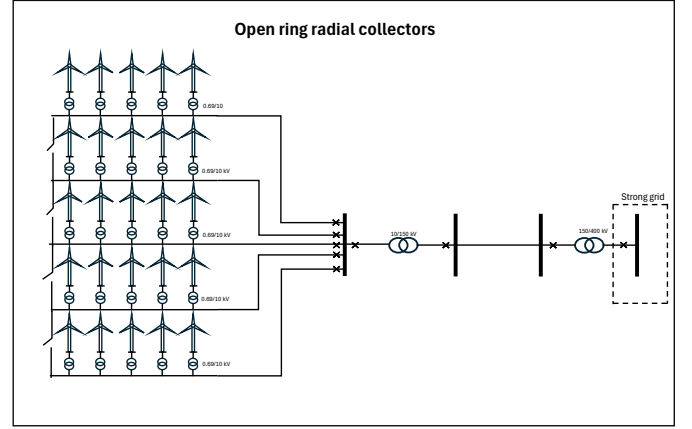


Figure 5. Electrical layout of the open ring radial collectors

layout [40].

Overall, the pure radial layout is attractive for a small project because of its low cost, simple design and easy operation, but it exposes the project to larger production losses when a feeder fault occurs. The open-ring radial layout introduces redundancy in the collection network. It improves reliability and operational flexibility and can increase the total energy yield over the project lifetime, but this comes at the price of higher investment and a more involved protection concept. The final choice between the two options should balance the benefit of improved availability against the additional cost and complexity.

In our case, an open-ring radial collector layout is considered the most reasonable solution, as it ensures higher redundancy. This will result in a higher investment cost for the electrical layout, however, we value a more robust system to minimize power production outages. From that perspective the open-ring radial collectors is the preferable solution for our wind farm.

IV. NETWORK INTEGRATION ISSUES

In this section, network integration issues will be explored. This includes calculating the maximum capacity of a wind farm that can be connected to different connection points and finding the most suitable connection points from a given layout, motivating the choice of connection point for our site and calculate the cost of connection.

A. Maximum connectable capacity

The network integration analysis was carried out by modelling each connection point as a purely inductive Thevenin equivalent X_{th} according to: [8]

$$X_{th} = \frac{U_n^2}{S_k},$$

where U_n denotes the nominal line-to-line voltage and S_k the short-circuit capacity. For each node, the minimum local load (where $S_{load,min}$ is apparent power and $P_{load,min}$ is active power) was determined from:[8]

$$S_{load,min} = n\sqrt{3}U_n I_{min}, \quad P_{load,min} = 0.9 S_{load,min},$$

using the minimum phase current $I_{\min}$ and number of parallel lines n from the given data. The active power transfer through an inductive link P is approximated by: [13]

$$P = \frac{U_s U_r}{X_{th}} \sin \delta,$$

and the conservative limit occurs at $\delta = 90^\circ$ with the receiving-end voltage at its upper bound $U_{\max}$. The resulting maximum transmittable power and hence the maximum connectable wind-farm power $P_{WF,max}$ are:

$$P_{WF,max} = P_{load,min} + \frac{U_s U_{\max}}{X_{th}}.$$

Applying these relations to the nine candidate buses yields the following approximate capacities (rounded to the nearest 10 MW):

Point	U_n [kV]	$P_{\max}(I_{\min})$ [MW]	$P_{\max}(I_{\max})$ [MW]
1	10	199.38	211.33
2	10	360.55	386.64
3	20	434.77	458.28
4	20	511.25	534.84
5	70	727.58	800.39
6	70	966.09	1001.40
7	200	1196.90	1267.30
8	200	1617.10	1687.80
9	380	20480	20480

The theoretical limits exceed the planned 80 MW by more than an order of magnitude. Consequently, economic rather than technical criteria dominate the connection decision.

B. Most suitable connection point

The most suitable case for our wind farm in terms of the possibility to install the wind farm as well as considering the economic aspects. It is assumed that the most economically efficient connection point is the point with the lowest short circuit capacity. If it is possible to connect to this point it will be the most suitable case. To be able to connect to the point, the voltage level have to stay between the range of $0.9U - 1.06U$ where U is the nominal voltage level [8]. A plot is created to easily visualize the change in voltage level depending on installed capacity. The calculations can be shown in Section IV-A.

Fig. 6. and Fig. 7. shows the voltage variation depending on the installed capacity. For the most economic case (point 1), the voltage level increases more than for the 10 kV bus. For our wind farm with a capacity of 80 MW, the voltage level for point 1 doesn't deviate much from the nominal voltage level, an increase of about 10%. This level is well within the acceptable range which makes the connection point the most suitable considering the low connection cost from the relatively low short circuit capacity. The case with the 10 kV bus is considered to visualize the results of connection points with higher short circuit capacity. Fig. 7 shows a less deviating voltage level for higher installed capacity. Therefore, we can assume that connection points with even higher short circuit capacity will keep the voltage level closer to the nominal voltage level. Therefore, it would be possible to connect the

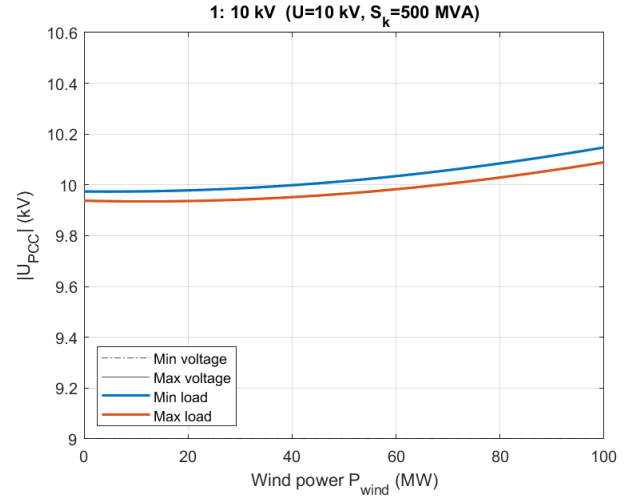


Figure 6. Voltage level of the 10 kV of Point 1 depending on the installed capacity

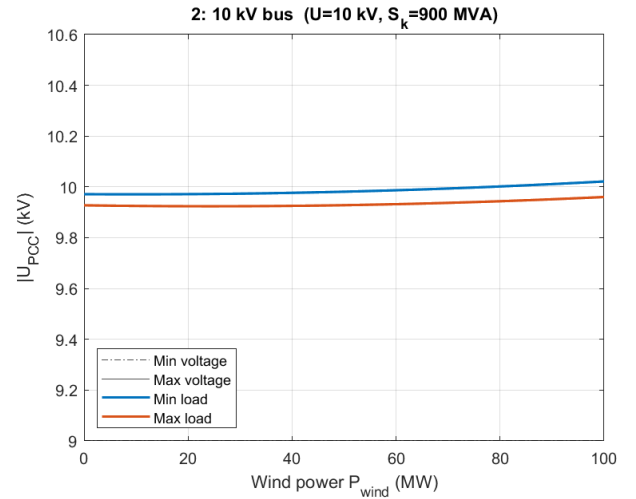


Figure 7. Voltage level of the 10 kV bus depending on the installed capacity

wind farm to each of the available connection points. From an economical perspective, connection point 1 is the most efficient point which will be most suitable for our wind farm.

C. Cost of connection

The project site is located on a ridge where the turbines are arranged in an arced, ridge-aligned pattern to reduce wake interaction. Each wind turbine operates at a generator voltage of 690 V and is equipped with a step-up transformer to 10 kV. The collection network therefore operates at 10 kV and the turbines are grouped into several radial feeders that converge at the on-site substation (average spur length per wind turbine is 300m).

The nearest grid node is Brattstrømmen at 150 kV, located approximately 30 km from the wind-farm substation (commissioning estimated for 2026) [35]. At the farm substation, the aggregated power is stepped up using a 10/150 kV main

transformer, and a single 150 kV export circuit connects to the actual utility network. Exporting at voltage U 150 kV can limit the current I to (where P is the rated power of the wind farm):

$$I = \frac{P}{\sqrt{3}U} \approx \frac{80 \text{ MW}}{\sqrt{3} 150 \text{ kV}} \approx 308 \text{ A.} \quad (9)$$

This minimizes resistive losses and voltage drop over the 30 km route and interfaces with a stiffer grid having higher short-circuit capacity.

To model the technoeconomic installation N denotes the number of turbines, L_{spur} the average medium-voltage (MV) spur length from each wind turbine to the wind farm bus collector (km), and L_{HV} the length of the 150 kV export line. The total unit costs are summarised in Table V.

Table V
UNIT COST ASSUMPTIONS ADOPTED [41].

Component	Unit Cost
MV overhead line (10 kV class)	$c_{\text{MV}} = 337,327 \text{ SEK/km}$
HV overhead line (123–170 kV)	$c_{\text{HV}} = 4,636,359 \text{ SEK/km}$
Transformer (120 MVA reference)	$C_{\text{trf},120} = 15,820,812 \text{ SEK/unit}$

The specific transformer cost c_{trf} becomes:

$$c_{\text{trf}} = \frac{15,820,812}{120 \text{ MVA}} \approx 131,840 \text{ SEK/MVA.}$$

This rating choice provides headroom for potential future expansion.

The cost components of the electrical infrastructure are computed using the following expressions. The MV collection network cost C_{MV} is

$$C_{\text{MV}} = N L_{\text{spur}} c_{\text{MV}},$$

the cost of the 150 kV overhead export line C_{HV} is

$$C_{\text{HV}} = L_{\text{HV}} c_{\text{HV}},$$

and the transformer cost C_{trf} is

$$C_{\text{trf}} = S_{\text{trf}} c_{\text{trf}},$$

where S_{trf} is the transformer rating in MVA. The total cost C_{tot} is therefore

$$C_{\text{tot}} = C_{\text{MV}} + C_{\text{HV}} + C_{\text{trf}} + C_{\text{sub}} + C_{\text{prot}},$$

where C_{sub} includes substation civil and electrical works (MV switchgear, HV bay, control and SCADA systems), and C_{prot} represents protection, metering and communication (SCADA) equipment.

For the selected installation, it is considered $N = 24$ turbines, an average spur length of $L_{\text{spur}} = 0.3 \text{ km}$, an export line of $L_{\text{HV}} = 30 \text{ km}$, and a main transformer rated at 120 MVA. The resulting costs are:

$$C_{\text{MV}} \approx 2.43 \times 10^6 \text{ SEK,}$$

$$C_{\text{HV}} \approx 1.391 \times 10^8 \text{ SEK,}$$

$$C_{\text{trf}} = 120131,840 \approx 1.58 \times 10^7 \text{ SEK}$$

Excluding substation and protection allowances, the partial total $C_{\text{tot,partial}}$ becomes

$$C_{\text{tot,partial}} \approx 1.57 \times 10^8 \text{ SEK}$$

It is assumed that some typical allowances for substation and protection works for the installation ($C_{\text{sub}} + C_{\text{prot}}$) to be 30 MSEK results in the total:

$$C_{\text{tot}} \approx 1.87 \times 10^8 \text{ SEK}$$

D. Connection point feasibility

When choosing a connection point it is not enough to look only at the maximum transferable power, voltage limits and line costs. One important additional issue is the expected level of curtailment. Different connection points may suffer from different levels of congestion in the existing grid. If the selected node is often heavily loaded, the system operator may have to curtail wind generation more frequently to respect thermal or stability limits. In that case a technically feasible and even cheap connection could still reduce the long-term energy yield and the income of the project. It is therefore important to consider present and planned power flows in the surrounding network and any known limitations in the operational practice of the TSO/DSO.

A second aspect is compliance with grid code requirements at the chosen voltage level [42]. Fault ride-through capability, reactive power range, voltage control and frequency support requirements may be stricter at higher voltage levels. Meeting these requirements can influence the design of the wind farm collector system, the choice of transformers and the control strategy of the turbines or the plant controller. Protection and fault clearing must also be coordinated with the existing relays and breakers at the connection point. For some nodes, additional protection equipment or communication links may be required, which increases cost and complexity and may cause delays during commissioning.

Reliability and future expansion also play a role. A connection at a higher voltage level may offer better redundancy in the transmission system and lower risk of long outages due to single line or transformer faults. The selected node should be able to accommodate possible repowering or extension of the wind farm during its lifetime without major reinforcements. It can also be advantageous to choose a point where planned grid upgrades are foreseen, since this may improve export capacity in the future. These aspects are not directly visible in a steady-state calculation but have a strong impact on the long-term performance of the project.

Finally, practical issues around construction, permitting and ownership must be considered. The route of the new line from the wind farm to the grid may cross environmentally sensitive areas, roads or villages, which can increase costs, cause delays, or even make some options infeasible. The division of responsibilities and connection charges between the wind farm owner and the grid operator can differ between voltage levels and between substations. All these non-technical factors need to be taken into account together with the electrical analysis when deciding on the most suitable connection point [43].

E. Reactive power affecting maximum connectable capacity

Allowing the wind farm to vary its power factor increases the amount of active power that can be transferred because of the voltage level can be held closer to the nominal voltage. This can be seen from the following approximation where the susceptance B is equal to zero [44]:

$$U_j = \frac{U_k^2 + 2(P_{kj}R + Q_{kj}X)}{U_k} = U_k + \frac{2(P_{kj}R + Q_{kj}X)}{U_k} \quad (10)$$

In this approximation, U_j is the voltage at the point of common connection where the wind farm is connected. U_k is the voltage at the other end of the line. P_{kj} and Q_{kj} is the active and reactive power from the wind farm. An increase in the generation results in an increase in P_{kj} and Q_{kj} . R and X is the resistance and reactance of the line [44]. Assume all parameters are constant, we can clearly see that for higher values of Q the voltage level will be higher if the reactance is positive. If the grid have inductive characteristic, the X is positive in this case. The wind farm is able to produce power with a power factor of 0.95 capacitive, resulting in a negative value of Q . This will reduce the value of the right term of the equation which leads to a voltage level closer to the nominal voltage, assuming that the term $P_{kj}R$ is larger than the $Q_{kj}X$. This will result in the possibility to install more wind power because the voltage level can be held within the allowed limits for higher power production.

Point	U_n [kV]	PF_{opt}	$P_{WF(I_{min})}$ [MW]	$P_{WF(I_{max})}$ [MW]
1	10	0.95	306.1	318.2
2	10	0.95	552.7	578.9
3	20	0.95	669.6	693.3
4	20	0.95	788.7	812.5
5	70	0.95	995.6	1067.3
6	70	0.95	1337.2	1372.5
7	200	0.95	1650.5	1720.6
8	200	0.95	2235.7	2306.2
9	380	0.95	20480	20480

F. Impedance of the line

The impedance of the line is estimated using the values provided in the project presentation [45]. The impedance of the line is estimated to around $Z = 0.1 + j0.3\Omega/\text{km}$. As we have a transformer close to the wind farm, our lines is assumed to be around 4 km to the transformer station. Other than that, it is the same case as in Section IV-B.

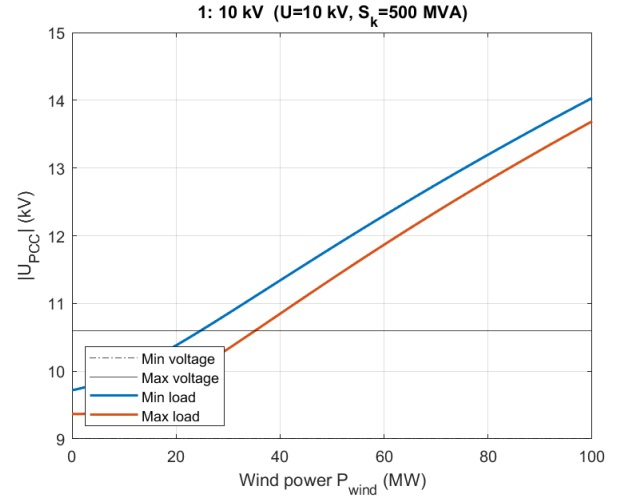


Figure 8. Voltage level of the 10 kV of Point 1 depending on the installed capacity, including the 4 km line

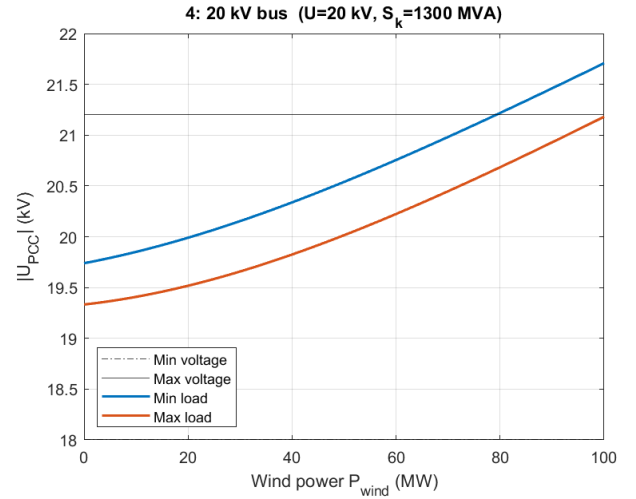


Figure 9. Voltage level of the 20 kV bus depending on the installed capacity, including the 4 km line

For the cases including the impedance of the line we can see a clear difference from our previous results. The 10 kV point that seemed to be a good connection point is now not feasible for our wind farm as the voltage rise is way higher than the allowed limit which can be seen in Figure 8. The Figure 9. shows that the voltage level reaches the allowed limit for the case with a minimum load in the system with the 20 kV bus. Therefore, this case is also not feasible for our wind farm. The voltage at the 70 kV side of the 20/70 kV transformer which can be seen in Figure 10. shows that the voltage level is not affected much for our wind farm size. Therefore, this would be the suitable point to connect our wind farm.

V. ECONOMIC ANALYSIS

In this section, the economic feasibility is discussed. This is done through estimating the future cash flow in and out of the

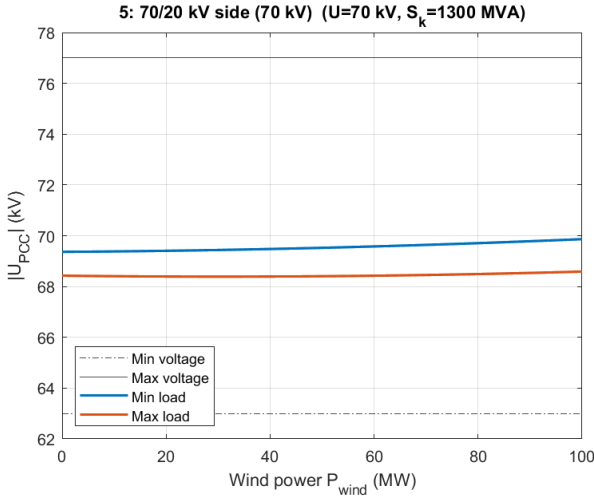


Figure 10. Voltage level of the 70 kV side of the 20/70kV transformer depending on the installed capacity, including the 4 km line

project, and calculating the net present value (NPV) and the Levelized Cost of Energy (LCOE) for a lifetime of 25 years.

A. Cashflow estimation

To estimate the cashflow in, some assumptions about future wholesale electricity prices were made. Based on a quantile prognosis of the day-ahead market price development until 2035 from [46], it is assumed that it is likely that the spot price in bidding area SE3 will increase slightly for at least the following decade. For this reason, the cashflow in is based on the average spot price in SE3 (P_0) according to this prognosis, which is around 50 €/MWh (approximately 512 SEK/MWh) and an slight yearly increase with an estimated annual growth rate (g) of 1% [46]. The expected wholesale price P_t in year t is therefore:

$$P_t = P_0(1 + g)^t. \quad (11)$$

Moreover, the capture rate of wind power is considered when estimating the cashflow in. Due to the so called cannibalization effect of renewable energies, wind and solar power production is negatively correlated with wholesale energy prices. The capture rate is a volume-weighted average price for a certain energy source [47]. In Sweden, where wind power penetration is high, the capture rate for wind power production in 2024 was around 70% [46]. This capture rate is estimated as rather stable, since capture rates of wind power production rarely falls below 70% even with penetration as high as a 50% share of total electricity production [47].

Considering the annual energy production AEP , the capture rate CR , and an average wholesale electricity price P , the total cashflow in CF_{in} at time t is:

$$CF_{in} = AEP \times CR \times P \times 1.01^t \quad (12)$$

The lifetime cost of a wind turbine include investment cost (which is assumed to cover the cost of connection), maintenance and operation costs, and dismantling costs. These are estimated based on previous reports and case studies.

The investment cost of a Vestas V136-3.45 MW turbine was estimated in a similar study to range between 952-1768 €/kW [48]. Our simulation results show that the NPV, LCOE, and overall project feasibility are highly sensitive to the assumed investment cost. Therefore, for the 80 MW wind farm, the lower and upper bounds derived from [48] are considered in the analysis. The corresponding total investment cost IC is given by:

$$IC_{upper} = 1768 \cdot 10^3 \text{ €/MW} \times 80 \text{ MW} = 141.44 \text{ M€} \quad (13)$$

$$IC_{lower} = 952 \cdot 10^3 \text{ €/MW} \times 80 \text{ MW} = 76.16 \text{ M€} \quad (14)$$

This corresponds to a total investment cost in the range of 76.16–141.44 M€, which is approximately 852–1584 MSEK

The external connection cost CC including lines, transformer, and substation and protection work was estimated in Section IV-C to 186 million SEK and it is assumed that is included in the investment cost.

The maintenance and operation cost (M&O) includes preventive and corrective maintenance, monitoring and control, financial management and site management. The cost is based on the article [49], where it is stated that the $M\&O$ has decreased from 17.5 €/MWh in 2005 to 13.2 EUR €/MWh in 2017. For this reason, the current M&O is estimated to approximately 10 €/MWh. The $M\&O$ cost each year is estimated as:

$$M\&O = AEP \times 10 \quad (15)$$

Lastly, the dismantling cost is estimated for a 2 MW turbine based on a real-life case in Gotland, Sweden from [50]. The cost of the crane and removal of the crane site was 575 000 SEK, and removal of the road was 50 000 SEK. The other costs of 500 000 SEK are assumed to be per turbine, including costs for work, transport and removal of the foundation. The total dismantling cost is then 40.6 million SEK.

The total cashflow out is then equal to the maintenance and operation cost each year plus the fixed investment and dismantling cost.

B. Net present value and LCOE

In order to estimate the net present value (NPV) and the Levelized Cost of Energy (LCOE), the cashflow out of the project must be defined. The lifetime cost of the wind farm consists of three components: the upfront investment cost (IC), the annual operation and maintenance cost ($M\&O_t$), and the dismantling cost (DC), which is assumed to occur in the final year of the project lifetime.

The total annual cash outflow $CF_{out,t}$ therefore becomes:

$$CF_{out,t} = \begin{cases} M\&O_t, & t = 1, \dots, T-1, \\ M\&O_t + DC, & t = T. \end{cases} \quad (16)$$

The net cashflow $CF_{net,t}$ in year t is then obtained as the difference between the cash inflow and outflow:

$$CF_{net,t} = CF_{in,t} - CF_{out,t}. \quad (17)$$

Table VI
MAIN ECONOMIC ASSUMPTIONS FOR THE WIND FARM PROJECT.

Project lifetime T	25 years
Discount rate r	6%
Installed capacity P_{rated}	80 MW
Annual energy production AEP	270297.6 MWh/yr
Initial wholesale price P_0	50 EUR/MWh $\approx$ 512 SEK/MWh
Annual spot price growth g	1% per year
Capture rate of wind CR	70%
Investment cost IC	0.852–1.58 BSEK
O&M cost $c_{O\&M}$	10 €/MWh $\approx$ 102 SEK/MWh
Dismantling cost DC	40.6 MSEK (paid in year 25)

For this specific project the values that are used for the estimation of the Net present value and LCOE are presented below:

The net present value (NPV) of the project is determined by discounting all future net cashflows using the discount rate $r = 6\%$ and subtracting the initial investment cost:

$$NPV = -IC + \sum_{t=1}^T \frac{CF_{\text{net},t}}{(1+r)^t}. \quad (18)$$

Finally, the Levelized Cost of Energy ($LCOE$) is computed as the ratio between the discounted lifetime costs and the discounted lifetime energy production:

$$LCOE = \frac{IC + \sum_{t=1}^T \frac{CF_{\text{out},t}}{(1+r)^t}}{\sum_{t=1}^T \frac{AEP}{(1+r)^t}}. \quad (19)$$

By applying (18) and (19) together with the economic assumptions summarised in Table VI, the two extreme investment cost cases lead to substantially different project feasibility.

The lower investment cost scenario, results in a positive net present value,

$$NPV_{\text{low}} = 109.11 \text{ MSEK},$$

and a comparatively competitive levelized cost of energy,

$$LCOE_{\text{low}} = 361.6 \text{ SEK/MWh}.$$

This indicates that, under favourable investment conditions, the project is economically viable and produces positive value over its lifetime.

For the high-cost boundary, the economic performance deteriorates significantly:

$$NPV_{\text{high}} = -622.02 \text{ MSEK},$$

$$LCOE_{\text{high}} = 573.2 \text{ SEK/MWh}.$$

In this case, the project becomes economically unattractive, since the net present value is strongly negative and the LCOE rises above typical market price levels.

These results clearly demonstrate that the economic feasibility of the project is strongly dependent on the assumed capital cost. Even within the realistic investment cost range reported in [48], the project's financial outcome shifts from distinctly profitable to clearly unprofitable. Moreover, when the high-investment cost case is considered, the resulting

LCOE exceeds the captured market price, indicating that the project would require either a long-term power purchase agreement (PPA) with a sufficiently high guaranteed price or an appropriate policy support mechanism (investment subsidies or a feed-in mechanisms) in order to achieve profitable operation.

VI. HYBRID SYSTEM

In the last part of the study a hybrid system with a Battery Energy Storage System (BESS) is investigated. The goal is to see if a battery can increase the revenue of the wind farm and reduce curtailment when solar power and a limited grid connection are present. Three optimisation problems are considered. First the yearly revenue of the wind farm is maximised without storage. Second the yearly revenue is maximised when a battery is installed and the investment cost of the battery is included. Third the total dispatched wind energy is maximised, without considering the electricity price, in order to minimise curtailment.

A. Modelling approach and constraints

The hybrid system is modelled as a single node that contains the 80 MW wind farm, a demand and a co-located solar plant. The node is connected to a stronger grid where both import and export are possible, but the transfer capacity is limited. Hourly data for one year is used and this year is assumed to be representative for the whole project lifetime of 25 years.

The wind power time series is based on the hourly power output of the selected Vestas V136-3.45 turbine at the chosen site from *ninja.renewables* [37], scaled to an installed capacity of 80 MW. The demand and solar time series come from the provided Swedish load and solar profiles for 2024. The load profile is multiplied by 1.2 times the wind farm capacity, which gives a peak demand of about 96 MW. The solar profile is multiplied by 0.5 times the wind farm capacity, which gives a peak solar output of about 40 MW.

Market prices are given by the hourly spot prices in bidding zone SE3 for 2024 taken from Nordpool [51]. All economic results in this section are expressed in Euro for one year.

The grid connection between the node and the main grid is restricted by a transfer limit equal to 25 percent of the wind farm rated power. The maximum import and export power is therefore 20 MW in every hour.

The Battery Energy Storage System is described by its energy capacity E_{max} in MWh and its charge and discharge power limits. In the optimisation model there are separate variables for charging power P_t^{ch} and discharging power P_t^{dis} for every hour t . Both are limited to a maximum of 40 MW. The state of charge E_t is bounded between 10 percent and 100 percent of the installed capacity. At the beginning of the year the state of charge is fixed at 50 percent of E_{max} .

The main constraints that appear in the model can be summarised as follows.

a) *State of charge balance*: For every hour after the first one the energy in the battery is updated by a simple balance

$$E_t = E_{t-1} + P_t^{\text{ch}} - P_t^{\text{dis}}. \quad (20)$$

For the first hour the state of charge is fixed to the chosen starting value. Charging and discharging efficiencies are assumed to be equal to one. This keeps the problem linear and easier to solve.

b) *State of charge limits:* For every hour the state of charge is bounded between a minimum level and the installed capacity,

$$0.1E_{\max} \leq E_t \leq E_{\max}. \quad (21)$$

These limits are implemented both as variable bounds and as explicit constraints. They prevent the model from overcharging the battery or fully depleting it.

c) *Power balance:* In each hour the sum of local generation and imports, minus the battery charging power and exports, must not exceed the demand at the node,

$$P_t^{\text{wind}} + P_t^{\text{solar}} + P_t^{\text{dis}} + P_t^{\text{imp}} - P_t^{\text{ch}} - P_t^{\text{exp}} \leq P_t^{\text{load}}. \quad (22)$$

Where P_t^{wind} and P_t^{solar} denotes the output power from wind and solar respectively, P_t^{imp} and P_t^{exp} denotes import and export power, and P_t^{load} denotes the demand. This inequality means that the model is not forced to meet the full demand. Instead it guarantees that the total power injected into the local network never exceeds what can be absorbed by the demand. When the available renewable generation and battery discharge are higher than the right hand side, the model reduces the dispatched wind power. The difference between the available wind potential and the dispatched wind power is interpreted as curtailed energy.

d) *Wind power limits:* For every hour the dispatched wind power cannot be higher than the available wind power $P_t^{\text{wind,avail}}$ from the time series and cannot be negative,

$$0 \leq P_t^{\text{wind}} \leq P_t^{\text{wind,avail}}. \quad (23)$$

e) *Import and export limits:* The imported and exported power are both limited by the grid connection,

$$0 \leq P_t^{\text{imp}} \leq P^{\text{imp,max}}, \quad 0 \leq P_t^{\text{exp}} \leq P^{\text{exp,max}}, \quad (24)$$

with $P^{\text{imp,max}} = P^{\text{exp,max}} = 20$ MW. In addition, the exported power cannot be larger than the available generation from wind, solar and the battery,

$$P_t^{\text{exp}} \leq P_t^{\text{wind}} + P_t^{\text{solar}} + P_t^{\text{dis}}. \quad (25)$$

This constraint avoids situations where the model would export more energy than it actually produces in that hour.

These constraints are used in the models with storage. In the model without storage the battery related variables and terms are removed.

Battery investment cost is described by a specific cost per installed MWh of energy capacity. Two values are considered, 300 000 and 450 000 per MWh. These values are chosen as a simple range that is consistent with recent large scale battery projects, for example the 300 MW and 650 MWh Mortlake battery in Australia [52]. The total investment cost of the battery is divided by a 25 year lifetime to obtain an equivalent annual cost. Degradation and fixed operation and maintenance of the battery are ignored.

The optimisation is implemented in Python using the linear solver HiGHS. Three separate optimisation problems are

formulated to represent the three objectives. The decision variables and constraints are the same, only the objective functions are changed.

B. Optimisation of revenue without BESS

In the first optimisation the wind farm operates without any storage. The decision variable is the dispatched wind power in each hour. It is limited by the available wind production, by the local demand and by the 20 MW export limit. Import is not used in practice because buying energy at a positive price does not increase the objective.

The objective function is to maximise the yearly revenue, $R_{\text{no BESS}}$:

$$R_{\text{no BESS}} = \sum_{t=1}^{8760} p_t P_t^{\text{wind}}, \quad (26)$$

where p_t is the spot price and P_t^{wind} is the dispatched wind power. The constraints that apply in this case are the nodal power balance without battery terms, the wind power limits and the import and export limits.

Under these conditions the optimal yearly revenue is about 8.11 million. This value is used as a reference for the cases with storage. It also represents the best that can be done if no battery is installed and if the operator can curtail wind power when there is congestion.

C. Optimisation of revenue with BESS

In the second optimisation a Battery Energy Storage System is added. The dispatch decisions now include the wind power, the battery charging and discharging power, and the import and export power. The constraints listed in Section VI.A apply in every hour.

The objective is to maximise the net yearly revenue including the battery, R_{BESS} . The revenue from selling wind power and discharging the battery is reduced by the cost of buying energy for charging. An annualised battery cost is then subtracted,

$$R_{\text{BESS}} = \sum_{t=1}^{8760} p_t (P_t^{\text{wind}} + P_t^{\text{dis}} - P_t^{\text{ch}}) - \frac{c_{\text{BESS}} E_{\max}}{25}, \quad (27)$$

where c_{BESS} is the specific investment cost per MWh.

For a specific cost of 450 000 per MWh the best result is obtained with a battery energy capacity of 45 MWh and a converter power of 40 MW. In this case the yearly net revenue increases to approximately 8.48 million. The gain compared to the case without storage is about 0.37 million per year, which corresponds to a relative increase of roughly 4.6 percent.

A sensitivity study is performed for a lower battery cost of 300 000 per MWh. For this cost level a larger storage of 133 MWh is tested, while keeping the same 40 MW power rating. The net yearly revenue in this case reaches about 8.97 million. The increase in net revenue compared to the reference without BESS is about 0.86 million per year, or around 10.6 percent. This shows that the economic benefit of storage is strongly dependent on the assumed investment cost.

D. Maximisation of power output

In the third optimisation the electricity price is removed from the objective. The model is instead set up to maximise the total dispatched wind power over the year, $E_{\text{wind,year}}$. The decision variables and constraints are the same as in the revenue maximisation with storage, but the battery investment cost is not included and the objective is purely technical.

The objective function is

$$E_{\text{wind,year}} = \sum_{t=1}^{8760} P_t^{\text{wind}}. \quad (28)$$

The solver finds, for each hour, the combination of wind dispatch, battery charging and discharging, and imports and exports that gives the highest yearly wind production subject to the network and storage limits.

When the 45 MWh battery is available, the optimisation result for this case gives a wind power output of about 265 710 MWh when the price profile is applied to the dispatch. For the larger battery with 133 MWh of storage the corresponding wind power output increases slightly to about 268 176 MWh.

E. Discussion of hybrid system performance

The optimisation results show that a moderate size Battery Energy Storage System can improve both the economic performance and the utilisation of the wind resource for the studied site. With a storage capacity of 45 MWh, which is slightly more than half an hour of full wind farm output, the yearly net revenue increases by about 4.6 percent when battery investment costs are taken into account. At the same time most of the otherwise curtailed energy during hours with high wind, high solar and limited export capacity can be shifted to periods with higher prices or higher local demand.

If future battery costs fall to around 300 000 per MWh, larger capacities become more attractive. The sensitivity case with 133 MWh shows a higher increase in yearly revenue and slightly lower curtailment. However, the additional gain in dispatched wind energy is small compared with the tripled storage capacity. This indicates that there are diminishing returns beyond a certain battery size for this specific system.

Overall, the hybrid wind plant with Battery Energy Storage offers a way to stabilise local supply and to reduce the loss of renewable energy caused by grid constraints and variable solar production. The results also underline that the economic feasibility of the battery depends strongly on technology cost assumptions and on future market conditions.

VII. DISCUSSION

The study set out to evaluate the technical and economic feasibility of an 80 MW onshore wind project in mountainous terrain in central Sweden, and the results show that such a project is in principle viable, but also highly sensitive to a number of design choices and cost assumptions. The most important cross cutting outcome is that technical feasibility is not the limiting factor for this site. Instead, economic performance and the specific characteristics of the grid connection dominate the overall assessment.

Regarding turbine technology and siting, the choice of the Vestas V136 with full converter generator can be seen as consistent with the combination of medium wind speeds, high turbulence and cold climate that characterise the Skärvagshållan area. The turbine meets the relevant wind class, offers a cold climate package and has a rotor size that remains manageable for transport and installation in complex terrain. At the same time, the validation against the Renewables.ninja benchmark showed that the combination of the manufacturer power curve and the Jensen wake model gives a capacity factor estimate that agrees very well with the independent reference. For the Enercon machine the necessary cubic approximation of the power curve leads to systematic overestimation of energy yield, which underlines how strongly turbine selection and data quality shape the credibility of the production assessment. In that sense the decision to proceed with a single turbine model that can be modelled with a realistic power curve reduces uncertainty in the subsequent grid and economic analysis.

The comparison of topological layouts further illustrates how aerodynamic performance must be balanced against the geometric constraints of the site. The arced ridge aligned configuration with 7D spacing yields slightly higher annual energy production and lower wake losses than the staggered layout, which is in line with expectations for turbines that follow the main wind direction along an exposed ridge. However, this configuration cannot be accommodated within the available area because it requires a much larger horizontal and vertical extent of approximately 13.9 km and 12.8 km. When the spacing along the arc is reduced to 4D in order to fit the terrain, wake losses increase and the annual energy production falls below the value obtained for the staggered layout. In practice this means that the theoretically attractive ridge aligned concept only becomes beneficial if sufficient land is available to maintain generous spacing between machines. For the present site, the two row staggered layout constitutes a pragmatic compromise, because it remains within the project boundaries while keeping wake related energy losses at a moderate level.

The network integration study emphasises that a purely capacity based view of the surrounding grid is not sufficient when deciding on the connection point. When line impedance is neglected, all nine candidate buses can easily accommodate the planned 80 MW plant and the lowest voltage point with the lowest short circuit capacity appears most attractive from an economic perspective because it suggests lower connection charges. Once realistic line impedance is included, the picture changes markedly. For the 10 kV point that initially seemed promising the voltage rise associated with the wind injection becomes much higher than the allowed limit and even the 20 kV bus reaches the upper voltage limit at minimum load. Only the 70 kV side of the 20/70 kV transformer shows a limited voltage deviation that remains compatible with the grid code for the considered power level. This outcome illustrates that the impedance of the connection line has a direct influence on the resulting voltage rise and can change the ranking of candidate nodes compared to an assessment that assumes zero impedance. Therefore, representing this impedance explicitly is essential when planning the connection of a new wind

project, even if the short circuit capacities at the surrounding nodes are high.

The analysis of reactive power capability confirms that the selected turbine concept can support higher connectable capacities because operation with a slightly capacitive power factor reduces the voltage rise at the point of common coupling. For the nine generic nodes the maximum admissible wind power increases significantly when a power factor of 0.95 is used instead of unity. However, in the present case the 80 MW plant remains far below these adjusted limits at all considered voltage levels. As a consequence, the reactive power flexibility of the farm improves the theoretical margin but does not constitute a binding constraint for the chosen connection arrangement.

In the economic assessment the most critical outcome is the pronounced sensitivity of net present value and levelised cost of energy to the assumed investment cost. Under the lower boundary of the cost range with $IC \approx 0.852$ BSEK the project yields a positive net present value of 109.11 MSEK and a levelised cost of energy of 361.6 SEK/MWh, which remains clearly below the assumed captured price. Under the upper boundary with $IC \approx 1.58$ BSEK the net present value becomes strongly negative with NPV of about minus 622.02 MSEK and the levelised cost rises to 573.2 SEK/MWh, which lies well above the captured market value. Given that the external grid connection alone contributes approximately 1.87×10^8 SEK, the results show that even moderate cost overruns relative to the optimistic case can push the project into an unprofitable regime. The use of a single representative capture rate of 70 percent and a modest annual price growth of 1 percent further means that the analysis does not explore more adverse price scenarios in which cannibalisation effects and slower price growth could deteriorate the economics even in the low cost case.

The hybrid system study adds another dimension to this picture by quantifying how storage can mitigate curtailment and improve revenue within a constrained grid connection. For the reference without battery, curtailment occurs whenever high wind and solar output coincide with limited export capacity of 20 MW and relatively low local demand. Introducing a battery of 45 MWh, which corresponds to slightly more than half an hour of full wind farm output, increases the yearly net revenue by about 4.6 percent even when the annualised battery investment cost is taken into account. This is achieved mainly by shifting otherwise curtailed energy from hours with high renewable production and low prices to periods with higher prices or higher local demand. A larger battery of 133 MWh yields a higher revenue increase of roughly 10.6 percent and slightly lower curtailment, but the additional gain in dispatched wind energy is small compared to the tripled storage capacity. This indicates that there is an economic optimum at moderate storage sizes for the given grid and price conditions and that further increases in capacity are subject to diminishing returns.

At the same time, the battery model relies on several simplifications. Degradation and fixed operation and maintenance costs for the storage system are neglected, the efficiencies are assumed to be unity and a single historical year is used as a proxy for the entire project lifetime with perfect foresight

of prices and renewable production. The grid connection is represented by a static transfer limit and the wider power system response to the hybrid plant is not modelled. These choices are reasonable for a first feasibility study, but they imply that the quantitative benefits of storage are likely to be somewhat overestimated and that the results should be interpreted as indicative rather than definitive.

Overall, the project demonstrates that a modern onshore wind farm in complex terrain near Idre can be technically integrated into the Swedish grid and can achieve competitive energy yields, provided that investment costs remain close to the lower end of the considered range and that an appropriate connection at higher voltage is chosen. The analysis also shows that storage can enhance the value of the project under constrained grid conditions, yet its deployment must be carefully sized in relation to battery costs and expected price patterns.

VIII. CONCLUSIONS

The work has presented an integrated assessment of an 80 MW wind project in a mountainous region in central Sweden, covering turbine technology, layout design, grid integration, economic performance and the potential role of battery storage. The main findings can be summarised along four dimensions.

First, with respect to technology and siting, the selected Vestas V136 full converter turbine combined with the Skärvagshållan site offers a technically consistent solution for a medium wind high turbulence environment with cold climate. The use of manufacturer power curve data and the validation against external production estimates support the reliability of the predicted annual energy production and capacity factor.

Second, the comparison of layouts shows that the two row staggered arrangement constitutes the most suitable configuration within the actual project boundaries. Ridge aligned arcs with 7D spacing can in principle yield slightly higher production with lower wake losses, but the required spatial extent exceeds the available area. If the spacing is reduced to 4D so that the arc fits the terrain, the advantage of the ridge alignment disappears and wake losses become larger than in the staggered case. This underlines that aerodynamic optimisation of layouts must always be evaluated together with land use constraints.

Third, the grid integration study indicates that the surrounding transmission system can accommodate the planned capacity, but that the choice of voltage level and the inclusion of realistic line impedance are decisive for voltage compliance at the point of connection. A naive assessment based solely on short circuit capacities would favour a low voltage node such as 10 kV because of lower assumed connection charges, yet the more detailed analysis reveals that voltage rise at 10 kV and 20 kV renders these options infeasible for the considered wind farm size once the line impedance is included. The 70 kV side of the 20/70 kV transformer emerges as the technically robust connection point, while the reactive power capabilities of the turbines provide additional margin but are not a limiting factor for the chosen configuration.

Fourth, the economic evaluation shows that the feasibility of the project is highly dependent on the specific investment cost. Under favourable cost assumptions with IC near 0.852 BSEK the project attains a positive net present value and a levelised cost of energy that is competitive relative to the expected captured price in the bidding zone. Under pessimistic cost assumptions with IC near 1.58 BSEK the project becomes clearly unprofitable, with a negative net present value and an LCOE above the captured market value. The hybrid system analysis demonstrates that a moderately sized battery of 45 MWh can increase yearly net revenue and reduce curtailment under a constrained 20 MW grid connection, yet also reveals that the benefits exhibit diminishing returns for larger storage capacities such as 133 MWh and remain sensitive to assumed battery prices and market conditions.

Taken together, these findings suggest that an onshore wind farm at Skärvagshållan with the chosen turbine technology and staggered layout is technically feasible and can be economically attractive if project costs are controlled and if a suitable high voltage connection is realised. Storage integration can further enhance revenue and resource utilisation in the presence of grid constraints, but should be designed cautiously and revisited as battery technology costs and electricity market structures evolve. Future work could refine the present analysis by applying more detailed flow modelling for complex terrain, exploring a broader range of price and policy scenarios and incorporating more comprehensive models of storage degradation and system interaction, in order to support an eventual investment decision on a firmer quantitative basis.

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